M-Tech Production & Marketing

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TASK 1 REPORT

Internship Program — Batch 2026

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1. Introduction

This report presents the work completed for Task Experiment 06 of the M-Tech Production & Marketing AI/ML Internship Program, Batch 2026. This task focuses on Unsupervised Learning, specifically exploring data clustering with K-Means and dimensionality reduction using Principal Component Analysis (PCA). M-Tech Production & Marketing, based in Haripur, KPK, offers this practical lab session to teach us how to find hidden structures, discover customer segments, and compress high-dimensional features without relying on pre-existing labels.

2. Objectives

- Understand Unsupervised Learning and how it works with unlabeled data.
- Learn to group data points into separate clusters using the K-Means algorithm.Master the Elbow Method to scientifically choose the perfect number of clusters (K).
- Apply Principal Component Analysis (PCA) to compress many features into two dimensions.
- Learn how to visualize clusters and reduced data components using charts.

3. Task Description

The task requires implementing clustering and dimensionality reduction using Scikit-Learn. We need to write Python code to generate synthetic data groups, apply the K-Means algorithm to find center points (centroids), and plot the error values (inertia) to find the "elbow" curve bend. Furthermore, we must take a multi-column dataset (like Iris), compress it using PCA down to two columns, and plot the output visually.

4. Work Done

I completed the lab exercise and successfully ran all five lab tasks inside a Jupyter Notebook. I generated blob data structures, calculated cluster locations, and used matplotlib to build clear scatter plots showing centroids. I also imported the Iris dataset, compressed its 4 features into 2 principal components while keeping around 96% of the original information, and mapped out a mock customer segmentation analysis.

5. Challenges Faced

Finding the True Elbow: Sometimes the error curve bends very smoothly, making it slightly tricky to pick the exact "elbow point" without testing multiple configurations.

Losing Information in PCA: Understanding that compressing features with PCA means losing a tiny bit of original information takes time to balance properly.

Plotting Dimensional Data: Visualizing raw data with 4 or more columns is impossible on a standard flat screen, which highlighted exactly why PCA is so helpful**.**

6. Key Learnings

- No Labels Needed: Unsupervised learning doesn't need answers/labels. It groups data entirely based on how close or similar the items are to each other.
- Centroid Behavior: K-Means works by putting down center markers called centroids and moving them around iteratively until the groups stabilize.
- PCA Utility: PCA solves the problem of "too many columns." It acts like a smart compression tool, merging data columns so we can easily plot high-dimensional items on simple 2D charts.

7. Conclusion

This lab provided a strong introduction to unsupervised learning workflows. By setting up K-Means clustering and PCA compression, I observed how machines extract hidden patterns entirely on their own. These techniques are highly useful in real-world business scenarios, such as separating online shoppers into clean target categories or speeding up massive machine learning models by removing redundant features.

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